PAPER_506: PI Infinity Decoder — Quantum State Phase Mapping

Unified Quantum Field Framework (UQFF)

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Source: PAPER_506_PI_Infinity_Decoder_Quantum_Mapping.md

PAPER_506: PI Infinity Decoder — Quantum State Phase Mapping

Session: 137 | **Source:** grok_share_84a767d3.txt (lines 3900–4310) **Date:** November 2025 — commit bc79f36 (PI_DIGITS_COUNT 312→728) **Related files:** source177_wolfram_field_unity.cpp (PI_Infinity_Decoder class)

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§1.1 Abstract

The PI Infinity Decoder maps the first 728 decimal digits of π (after the decimal point) into a quantum amplitude array of size `PI_DIGITS_COUNT × 1` via iterative phase accumulation. Each element of the array encodes a magnetic field amplitude that depends on digit value, sacred time constants, and quantum state index. The mapping allows any quantum state `i ∈ [0, 312)` to be assigned a unique, deterministic magnetic field value and a complex-valued DPM pair (UA' + i·SCm) derived from π's infinite non-repeating sequence.

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§1.2 Array Construction

```
PI_DIGITS_COUNT = 728 = 26 × 28     (26D UQFF × 28 extended sacred multiplier)
QUANTUM_STATES  = 26                 (one per UQFF dimension)

pi_digits[728] = { 1,4,1,5,9,2,6,5,3,...  }   (first 728 post-decimal digits of π)

Phase accumulation:
  phase_0 = 0
  phase_i = phase_{i-1} + pi_digits[i] × (π/7)    (INFINITY_RATIO = π/7)

Magnetic field amplitude:
  A_i = sin(2π × phase_i) × (1 + cos(phase_i × f_Schumann))
```

```
where f_Schumann = 7.83 Hz (Schumann resonance)
```

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§1.3 getMagneticField(state, time_phase)

```
B(state, t) = A_{state mod 728} × sin(t × ϕ / T_Baktun)

where:
  ϕ         = 1.6180339887  (golden ratio)
  T_Baktun  = 144000.0      (Mayan Baktun in days)
  state     ∊ [0, QUANTUM_STATES-1]
```

Physical interpretation: The Mayan Baktun period (394.26 years ≈ 144,000 days) acts as the time normalizer for the magnetic orbit equation. The golden ratio modulates how rapidly adjacent states evolve. This produces a deterministic but quasi-random field pattern across all 26 quantum states at any given time.

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§1.4 getDPM_Pair(state) — Complex Plane Encoding

```
DPM_pair(state) = A_{state} + i × A_{(state+13) mod 728}

Real part  = UA&#39; component (active, measured)
Imaginary  = SCm component (superconductive, virtual)
13-offset  = half of 26 UQFF dimensions = counter-phase partner
```

This maps the di-pseudo-monopole pair (UA', SCm) directly from the π digit sequence, providing an infinite, non-repeating source of field values grounded in the mathematical constant π.

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§1.5 getConsciousnessResonance(lineage_level)

The 7 sacred time constants act as phase modulators in a 7-term co-sum:

```
R(■) = (1/7) × Σ_{k=1}^{7} f_k(■)

where:
  f_1(■) = sin(■ × T_gen)         T_gen     = 40.0 years (Biblical generation)
  f_2(■) = cos(■ × T_katun)       T_katun   = 7200.0 days (Mayan Katun)
  f_3(■) = sin(■ × T_tun)         T_tun     = 360.0 days (Mayan Tun)
  f_4(■) = cos(■ × ϕ)             ϕ         = 1.6180339887 (golden cycle)
  f_5(■) = sin(■ × f_Sch)         f_Sch     = 7.83 Hz (Schumann resonance)
  f_6(■) = cos(■ × 7.83)          (Schumann second application)
  f_7(■) = sin(■ × (π/7))         INFINITY_RATIO

This is a 7-linear-independent-frequency co-sum, orthogonal by construction.
```

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§1.6 PI_DIGITS_COUNT Expansion (312 $\rightarrow$ 728)

The original implementation used `std::array<int, 312>` (= 26 × 12). The initializer list contained more than 312 elements causing MSVC error C2078. The fix was:

```
constexpr int PI_DIGITS_COUNT = 728;   // 26 × 28 (next sacred integral multiple)
constexpr std::array&lt;int, PI_DIGITS_COUNT&gt; pi_digits = { ... };  // 728 elements
static_assert(pi_digits.size() == PI_DIGITS_COUNT, &quot;PI digits mismatch&quot;);
std::array&lt;double, PI_DIGITS_COUNT&gt; infinite_curve;  // matched size
```

§1.7 Equations Summary

```
Phase function:    ϕ_i = Σ_{j=0}^{i} d_j × (π/7),              d_j ∊ {0,...,9}
Amplitude:         A_i = sin(2π ϕ_i) × (1 + cos(ϕ_i × 7.83))
Mag field:         B(s,t) = A_{s mod 728} × sin(t × ϕ_gold / 144000)
DPM pair:          DPM(s) = A_s + i × A_{(s+13) mod 728}
Resonance:         R(■) = ¹/■ Σ_{k=1}^{7} fk(■)                 [dimensionless]
```

§1.8 Citation

Source: grok_share_84a767d3.txt, lines 3900–4310 (source177 full code) Commit: bc79f36 — "source177 PI_DIGITS_COUNT update" Related: PAPER_508 (Sacred Time Constants) Paper number: PAPER_506